

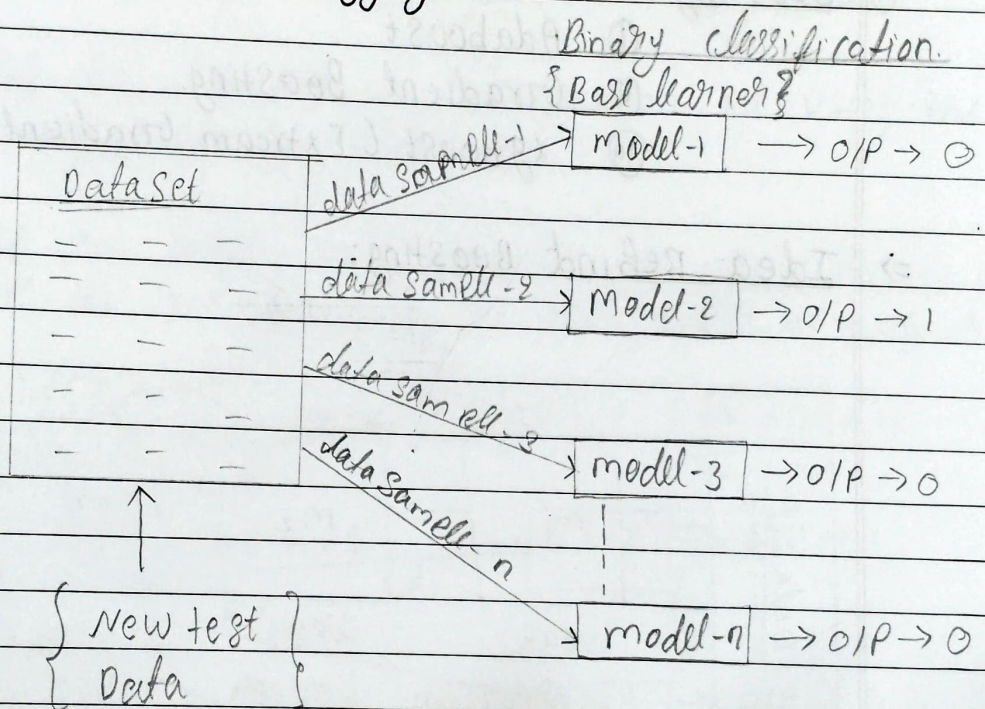
Ensemble Techniques

⇒ Bagging And Boosting

① Bagging:

① Random forest.

⇒ Idea behind Bagging.



⇒ In this case your models are same or different algorithms.

It is give the output based on the majority voting classifier.

In the given situation the prediction of Ensemble model is 0 because this is majority output.

In case of Regression, it is taking the avg of all the output.

{ Note: One most important thing the models or base-Learners are trained Parallyly.

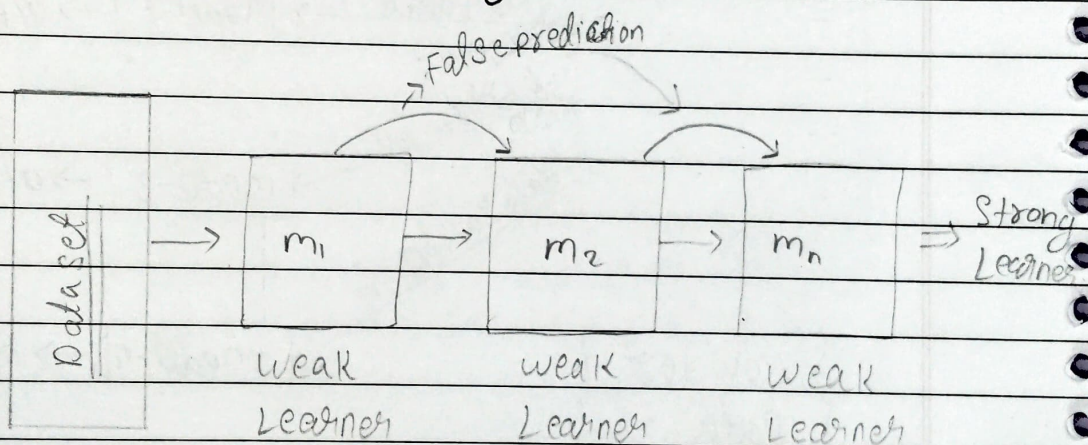
(i) Boosting:

(i) Adaboost

(ii) Gradient Boosting

(iii) Xgboost (Extream Gradient Boosting.)

⇒ Idea Behind Boosting:



This model also known as sequential model because, it is train the model One by one on the taken some data from the dataset and some from previous models miss classifying data.

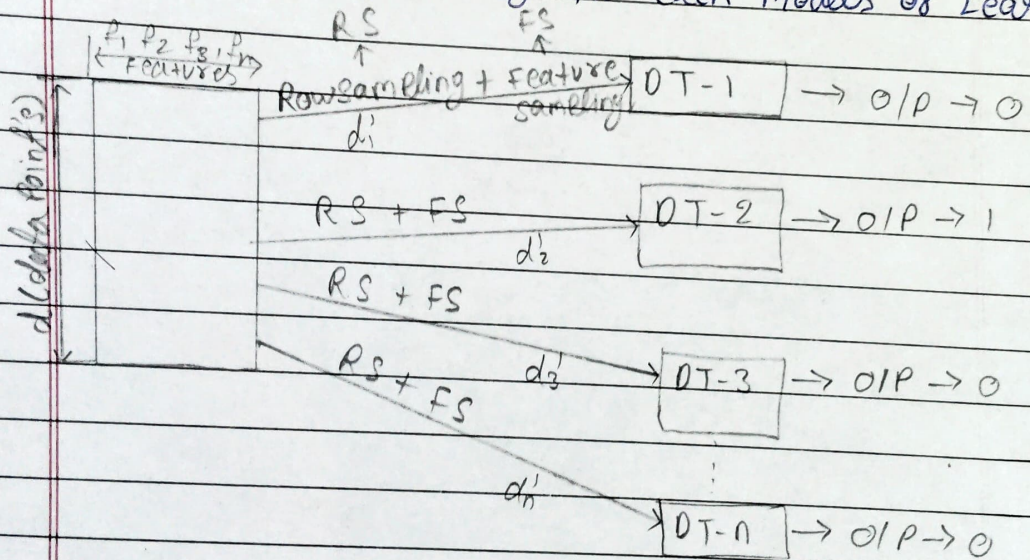
The base models are weak learners and after one-another model training at the end it become a strong learner.

Random Forest Classification And Regression:-

This machine learning algorithm falls under the Bagging.

This is use the Decision Tree algorithm as its base Learner. and train that base Learners on the some (Row sample + Features sample) data and give the outputs on the new dataset.

- ⇒ If the Problem is classification, it using the majority voting classifier.
- ⇒ If the Problem is Regression, it using the avg of all the O/P given by the each models or Learner.



Q. Why we use the Random forest, when we have Decision Tree?

we all are want a Generalised model which have "low bias and low variance."

Decision tree



overfitting the model



(i) Low bias

(ii) High Variance

Random Forest



Generalised model



(i) Low Bias

(ii) Low variance.

by using the Random forest or multiple Decision tree on the different data points it achieve the Low Bias and low variance.